HW3

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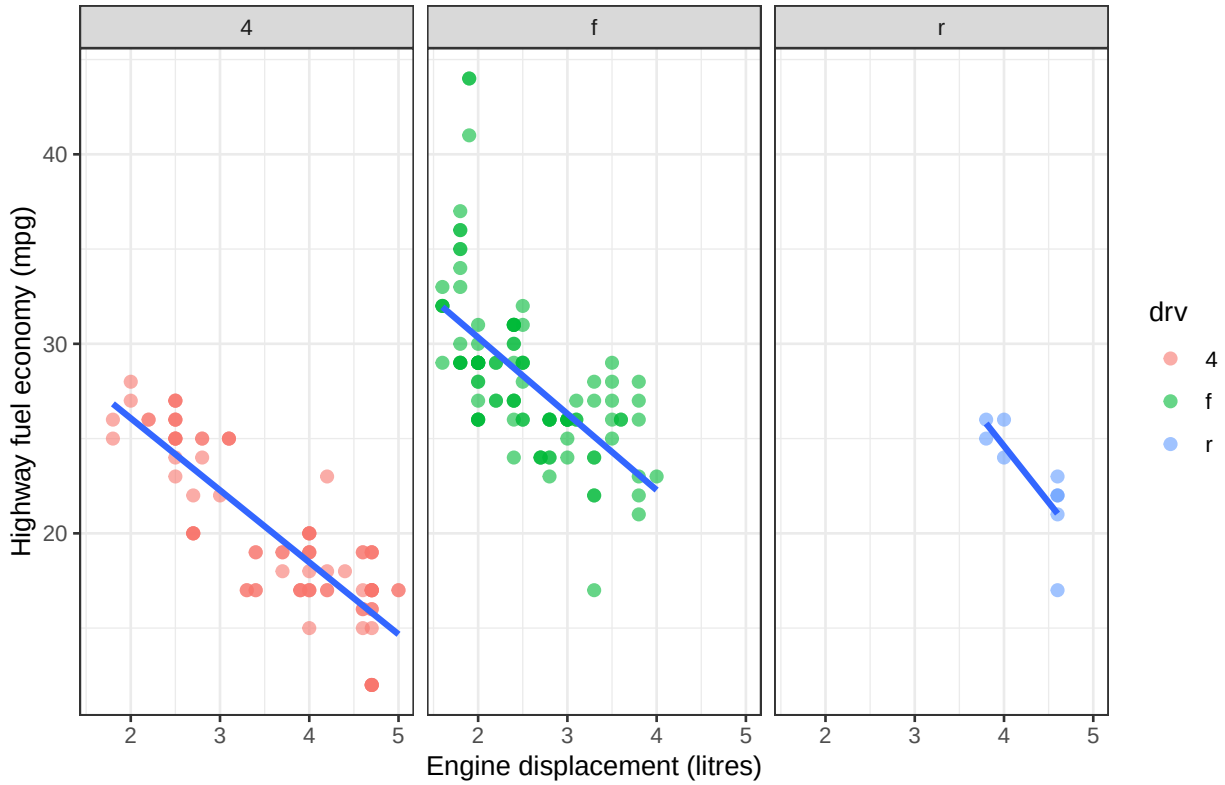
2026-11-04

```r
# GitHub repository: <PASTE YOUR REPO LINK HERE>
# install.packages(c("ggplot2", "dplyr", "tidyr", "readr", "sf"))
library(ggplot2)
library(dplyr)
library(tidyr)
library(readr)
library(sf)
vehicles <- ggplot2::mpg # Full dataset
gems <- ggplot2::diamonds # Full dataset

#Problem 1
#Part A
vehicles_a <- vehicles %>%
  filter(displ <= 5)

plot_1a <- ggplot(vehicles_a, aes(x = displ, y = hwy)) +
  geom_point(aes(color = drv), size = 2, alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x,
              color = "#3366FF", linewidth = 1.2) +
  facet_wrap(~ drv) +
  labs(
    x = "Engine displacement (litres)",
    y = "Highway fuel economy (mpg)",
    color = "drv"
  ) +
  theme_bw()

plot_1a
```

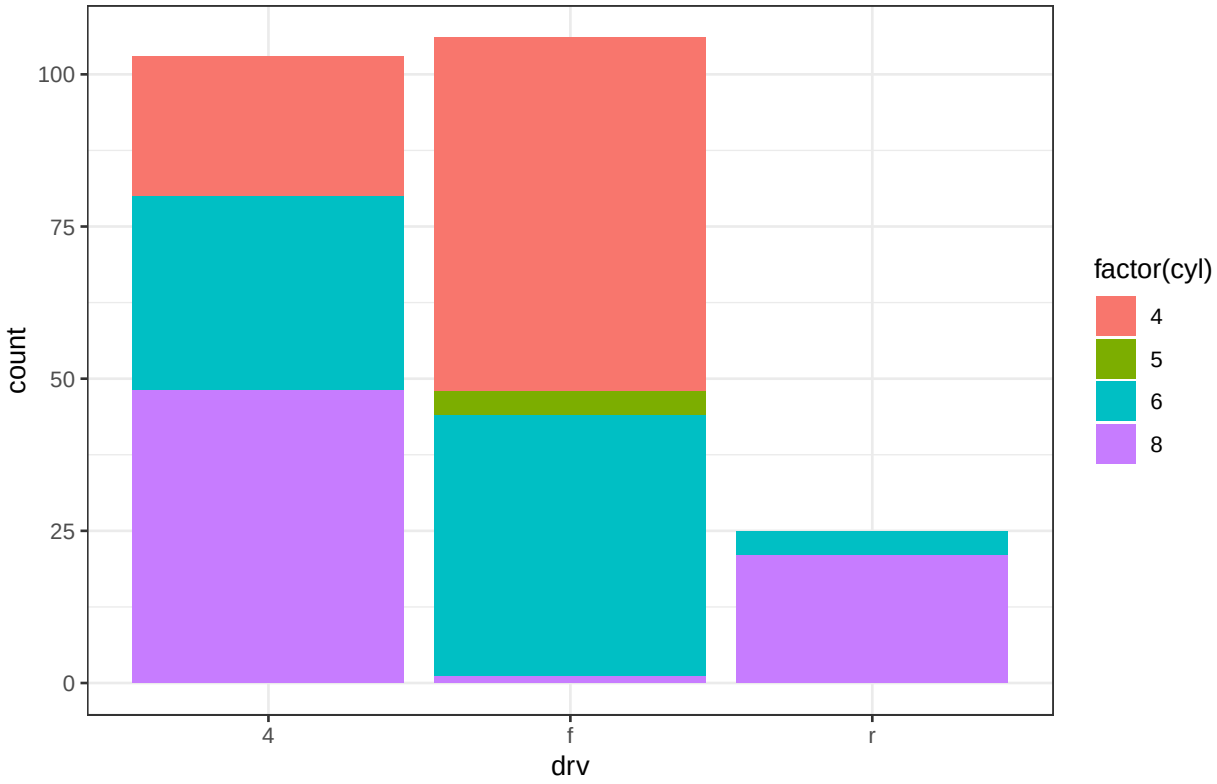



```r
ggsave("plot_1a.png", plot_1a, width = 9, height = 4.5, dpi = 150)

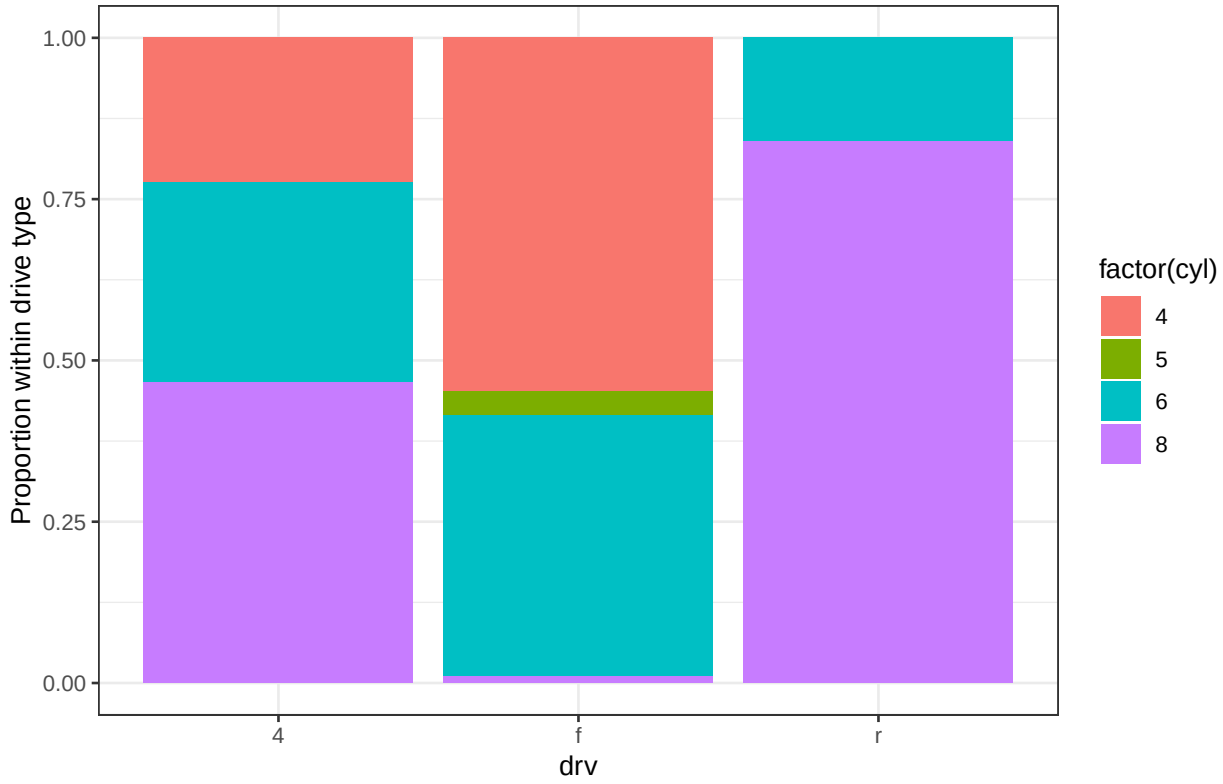
#Second part
# Because of facet_wrap(~ drv). Faceting splits the data into one subset per
# value of drv (4, f, r) and draws EVERY layer separately in each panel. So
# geom_smooth() gets its own subset in each panel and fits a separate
# regression of hwy on displ for that drive type: 3 panels = 3 lines.
# The color = drv mapping in geom_point() does not cause this. It is local to
# the point layer, so it colors the points and creates the legend, but
# geom_smooth() never sees it. Without the facet (all cars in one panel),
# geom_smooth() would have no grouping variable and would draw just ONE line
# through all the cars. The lines are all the same blue because the smooth
# layer uses a fixed color, not a mapped one.

#Part B

ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
  geom_bar() +
  theme_bw()
```




```
#Second part
ggplot(vehicles, aes(x = drv, fill = factor(cyl))) +
  geom_bar(position = "fill") +
  labs(y = "Proportion within drive type") +
  theme_bw()
```




```
# position = "fill" rescales every bar to add up to 1, so each bar shows the share
# of each cylinder count, not the number of cars. All bars are the same height no
# matter how many cars they hold. The count plot shows the real sizes differ:
# 103 four-wheel drive, 106 front-wheel drive, but only 25 rear-wheel drive cars.

#Part C
max_carat <- 3   # smaller table: diamonds of 3 carats or less

gems_small <- gems %>%
  filter(carat <= max_carat)

n_total    <- nrow(gems)
n_retained <- nrow(gems_small)
n_excluded <- n_total - n_retained
cat("Original records:", n_total, "\n")
```

```
## Original records: 53940
```

```
cat("Retained (carat <=", max_carat, "):", n_retained, "\n")
```

```
## Retained (carat <= 3 ): 53908
```

```
cat("Excluded (carat >", max_carat, "):", n_excluded, "\n")
```

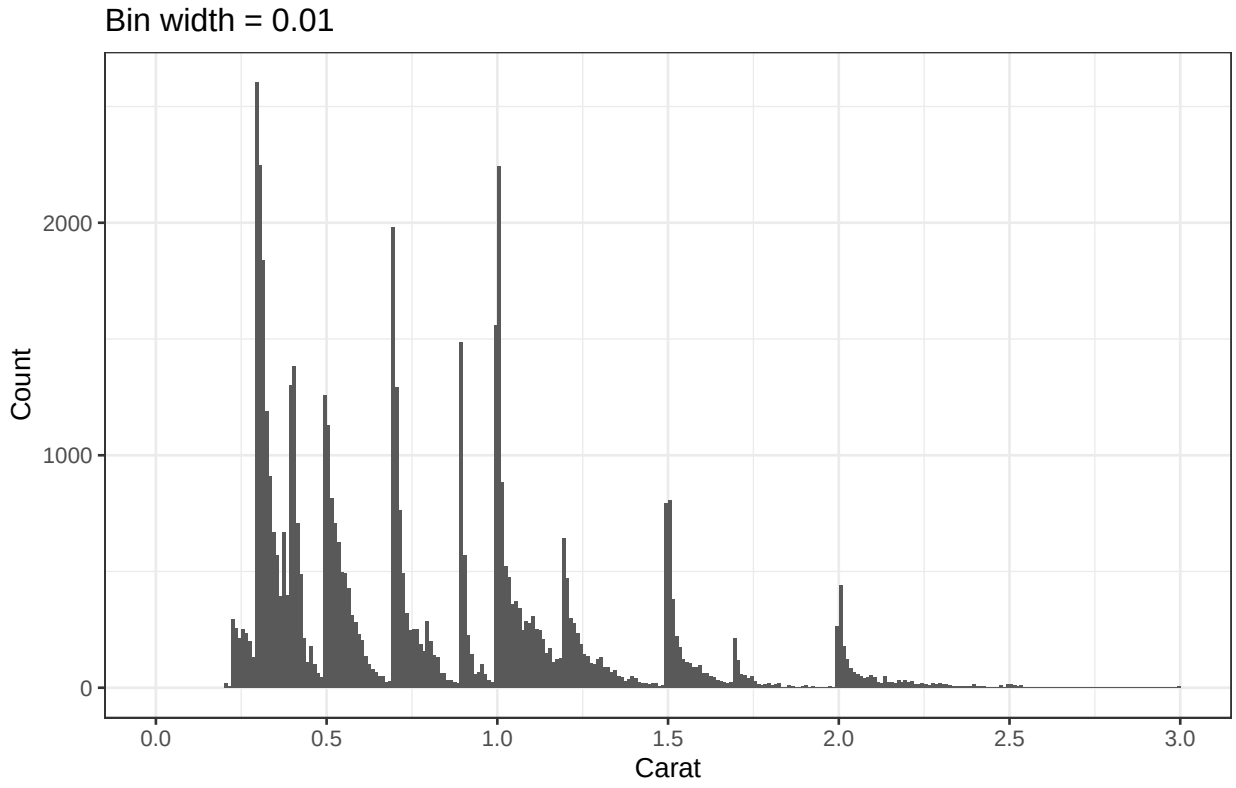
```
## Excluded (carat > 3 ): 32
```

```
p_fine <- ggplot(gems_small, aes(x = carat)) +
  geom_histogram(binwidth = 0.01, boundary = 0, closed = "right") +
  coord_cartesian(xlim = c(0, max_carat)) +
```

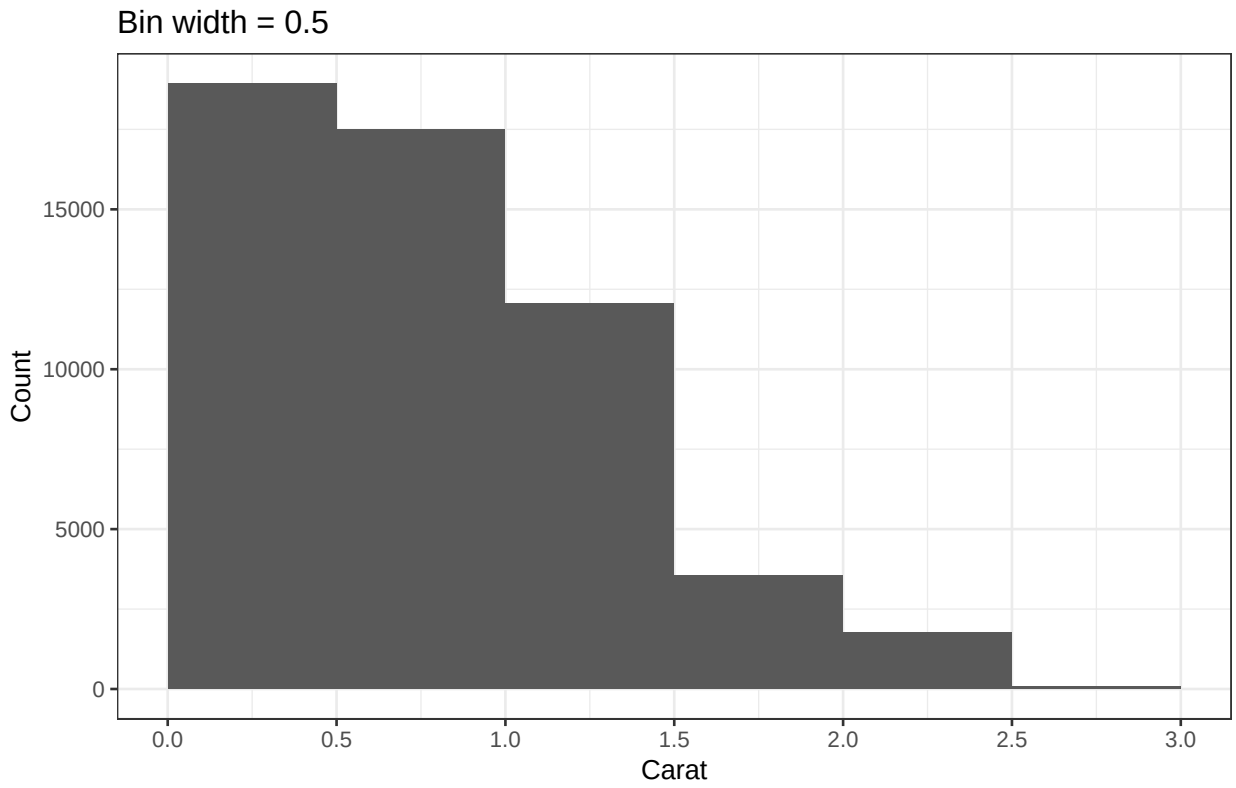
```
  scale_x_continuous(breaks = seq(0, max_carat, by = 0.5)) +
  labs(title = "Bin width = 0.01", x = "Carat", y = "Count") +
  theme_bw()

p_coarse <- ggplot(gems_small, aes(x = carat)) +
  geom_histogram(binwidth = 0.5, boundary = 0, closed = "right") +
  coord_cartesian(xlim = c(0, max_carat)) +
  scale_x_continuous(breaks = seq(0, max_carat, by = 0.5)) +
  labs(title = "Bin width = 0.5", x = "Carat", y = "Count") +
  theme_bw()

p_fine
```




```
p_coarse
```




```
ggsave("p_fine.png", p_fine, width = 7, height = 4, dpi = 150)
ggsave("p_coarse.png", p_coarse, width = 7, height = 4, dpi = 150)

# Part 2

gems %>%
  filter(round(carat, 2) %in% c(0.99, 1.00)) %>%
  count(carat)
```

```
## # A tibble: 2 x 2
##   carat     n
##   <dbl> <int>
## 1  0.99    23
## 2  1     1558
```

```
#   carat     n
#    0.99    23
#    1.00  1558
# Only 23 diamonds are recorded at 0.99 carat, but 1,558 at exactly 1.00.

# Records retained and excluded (from the cat() output above):
#   Original gems:            53,940 records
#   Retained (carat <= 3):    53,908 records
#   Excluded (carat > 3):         32 records (very large stones in the far
#                                 right tail, which would stretch the x-axis)

# Part 3
# With 0.01-carat bins, the distribution shows sharp spikes at "round"
```

```
# weights such as 0.3, 0.5, 0.7, 0.9, 1.0, 1.5 and 2.0 carats, and each
# spike is followed by a gradual decline in counts just above it. Very few
# diamonds sit just below these weights (23 at 0.99 vs. 1,558 at 1.00),
# suggesting stones are cut or recorded to reach the popular sizes. The
# 0.5-carat bins merge all of this into a smooth right-skewed shape, so the
# spikes are invisible.

#Problem 2
data_sources <- tibble(
  dataset = c("Libraries - Locations, Contact Information, and Usual Hours of Operation",
              "Boundaries - Community Areas"),
  publisher = c("City of Chicago Data Portal (Chicago Public Library)",
                "City of Chicago Data Portal"),
  url =
  ↪  c("https://data.cityofchicago.org/Education/Libraries-Locations-Contact-Information-and-Usual-/x8

          ↪  "https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Community-Area
  format = c("CSV", "GeoJSON"),
  accessed = c("2026-11-04", "2026-11-04")
)
write_csv(data_sources, "data_sources.csv")

libraries_raw <-
↪  read_csv("~/Downloads/Libraries_-_Locations___Contact_Information__and_Usual_Hours_of_Operation.csv
↪  show_col_types = FALSE)
areas <- st_read("~/Downloads/Boundaries - Community Areas.geojson", quiet = TRUE)
nrow(libraries_raw)
```

```
## [1] 82
```

```
libraries <- libraries_raw %>%
  extract(LOCATION, into = c("lat", "lon"),
          regex = "\\(([-0-9.]+),\\s*([-0-9.]+)\\)", convert = TRUE) %>%
  rename(facility_name = BRANCH, address = ADDRESS)

sum(is.na(libraries$lat) | is.na(libraries$lon))
```

```
## [1] 0
```

```
range(libraries$lat)
```

```
## [1] 41.65720 42.00682
```

```
range(libraries$lon)
```

```
## [1] -87.8142 -87.5336
```

```
sum(duplicated(libraries$facility_name))
```

```
## [1] 0
```

```
sum(duplicated(libraries$address))
```

```
## [1] 0
```

```r
sum(duplicated(libraries[, c("lat", "lon")]))
```

```
## [1] 0
```

```r
libraries_clean <- libraries %>%
  filter(!is.na(lat), !is.na(lon)) %>%
  distinct(facility_name, address, .keep_all = TRUE) %>%
  mutate(site_id = sprintf("LIB%03d", row_number())) %>%
  select(site_id, facility_name, address, lat, lon)
nrow(libraries_clean)
```

```
## [1] 82
```

```r
sites_m <- st_as_sf(libraries_clean, coords = c("lon", "lat"), crs = 4326) %>%
  st_transform(26971)
areas_m <- st_transform(areas, 26971)
st_crs(sites_m)$Name
```

```
## [1] "NAD83 / Illinois East"
```

```r
st_crs(sites_m)$units_gdal
```

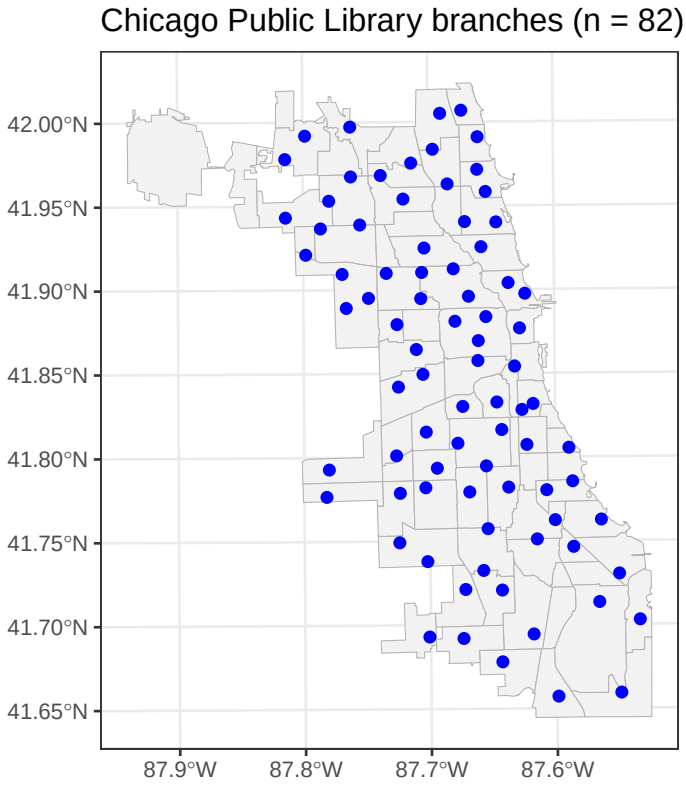
```
## [1] "metre"
```

```r
# Maps
buffer_500  <- st_union(st_buffer(sites_m, dist = 500))
buffer_1000 <- st_union(st_buffer(sites_m, dist = 1000))

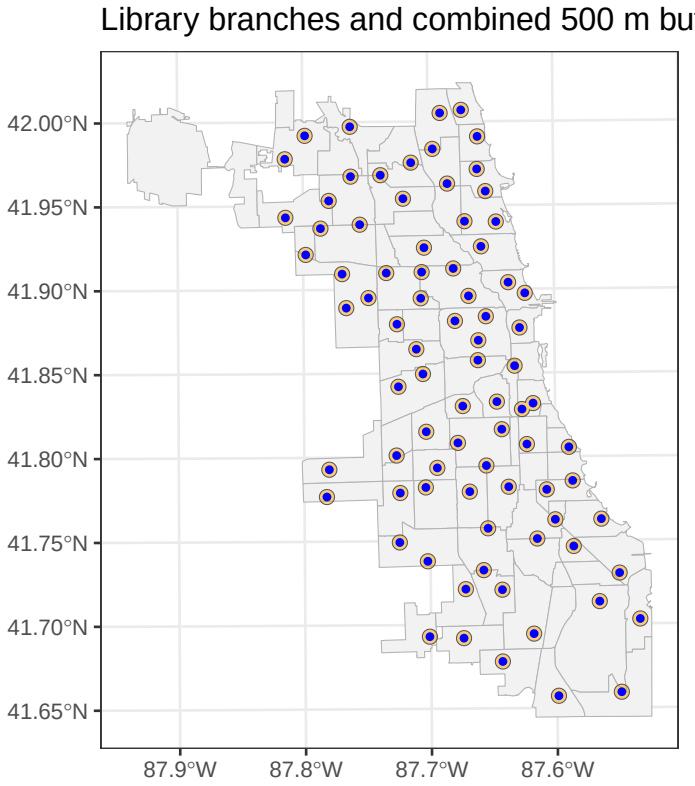
p_sites <- ggplot() +
  geom_sf(data = areas_m, fill = "grey95", color = "grey70") +
  geom_sf(data = sites_m, color = "blue", size = 1.8) +
  labs(title = "Chicago Public Library branches (n = 82)") +
  theme_bw()

p_buffers <- ggplot() +
  geom_sf(data = areas_m, fill = "grey95", color = "grey70") +
  geom_sf(data = buffer_500, fill = "orange", alpha = 0.5) +
  geom_sf(data = sites_m, color = "blue", size = 1) +
  labs(title = "Library branches and combined 500 m buffer") +
  theme_bw()

p_sites
```


Chicago Public Library branches (n = 82)
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W

```
p_buffers
```


Library branches and combined 500 m buffer
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W

```r
# Areas
buffer_summary <- tibble(
  radius_m = c(500, 1000),
  union_area_km2 = round(as.numeric(c(st_area(buffer_500), st_area(buffer_1000))) / 1e6,
  ↪  2)
)
buffer_summary
```

```
## # A tibble: 2 x 2
##   radius_m union_area_km2
##      <dbl>          <dbl>
## 1      500           64.3
## 2     1000          250.
```

```r
# Data checks:
# Facilities are Chicago Public Library branches in the City of Chicago.
# Source is recorded in data_sources.csv; the raw files are not changed.
# Records before cleaning: 82. Records after cleaning: 82. No records removed.
# Coordinates: none missing, all within Chicago (lat 41.66 to 42.01, lon -87.81 to
↪  -87.53).
# Duplicates: no repeated names, addresses, or coordinates.
# Each branch has a unique ID from LIB001 to LIB082.
# CRS: EPSG:26971 (NAD83 / Illinois East State Plane). Units: meters.

# Area table:
#   radius_m  union_area_km2
#        500           64.3
#       1000          250.0

# Interpretation:
# Branches are spread fairly evenly, about 1.9 km apart on average, so the 500 m
# buffers barely overlap and together cover only about 11% of the city. Doubling the
# radius to 1,000 m roughly quadruples the covered area to about 42% of the city.
# The biggest gaps are on the far Northwest Side and in some South Side and industrial
↪  areas.

#Problem 3
districts <- areas_m %>%
  transmute(district_id = as.integer(area_numbe),
            district_name = community)

nrow(districts)
```

```
## [1] 77
```

```r
sum(is.na(districts$district_id))
```

```
## [1] 0
```

```r
sum(duplicated(districts$district_id))
```

```
## [1] 0
```

```r
sum(st_is_empty(districts))
```

```
## [1] 0
```

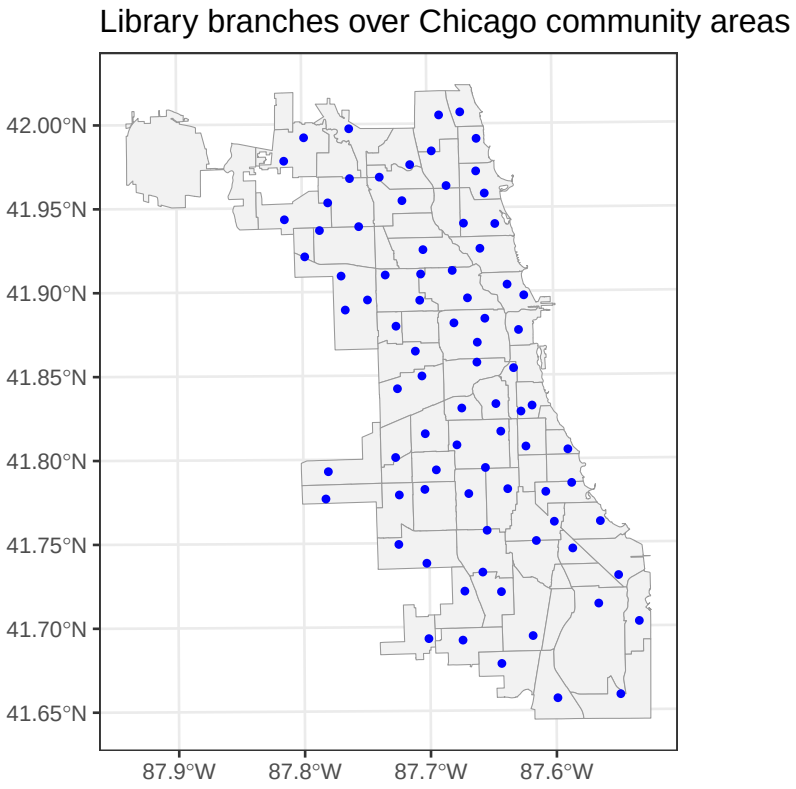
```
sum(!st_is_valid(districts))
```

```
## [1] 0
```

```
st_crs(districts) == st_crs(sites_m)
```

```
## [1] TRUE
```

```
ggplot() +
  geom_sf(data = districts, fill = "grey95", color = "grey60") +
  geom_sf(data = sites_m, color = "blue", size = 1) +
  labs(title = "Library branches over Chicago community areas") +
  theme_bw()
```




```
# Matches and summary
site_match_audit <- tibble(
  site_id = sites_m$site_id,
  n_matches = lengths(st_intersects(sites_m, districts))
)
write_csv(site_match_audit, "site_match_audit.csv")
table(site_match_audit$n_matches)
```

```
##
##  1
## 82
```

```
site_match_audit %>% filter(n_matches != 1)
```

```
## # A tibble: 0 x 2
## # i 2 variables: site_id <chr>, n_matches <int>
```

```
assigned <- sites_m %>%
  filter(site_match_audit$n_matches == 1) %>%
  st_join(districts, join = st_intersects)
n_assigned <- nrow(assigned)
n_unassigned <- nrow(sites_m) - n_assigned

district_counts <- districts %>%
  left_join(count(st_drop_geometry(assigned), district_id, name = "n_sites"),
            by = "district_id") %>%
  mutate(n_sites = replace_na(n_sites, 0),
         area_km2 = as.numeric(st_area(geometry)) / 1e6,
         sites_per_km2 = n_sites / area_km2)

all(district_counts$area_km2 > 0)
```

```
## [1] TRUE
```

```
sum(district_counts$n_sites) == n_assigned
```

```
## [1] TRUE
```

```
sum(district_counts$n_sites == 0)
```

```
## [1] 16
```

```
district_counts %>%
  st_drop_geometry() %>%
  as_tibble() %>%
  arrange(desc(n_sites)) %>%
  print(n = 77)
```

```
## # A tibble: 77 x 5
##    district_id district_name          n_sites area_km2 sites_per_km2
##          <int> <chr>                    <int>    <dbl>         <dbl>
##  1          15 PORTAGE PARK                 3    10.2          0.293
##  2          23 HUMBOLDT PARK                3     9.34         0.321
##  3          25 AUSTIN                       3    18.5          0.162
##  4          28 NEAR WEST SIDE               3    14.7          0.204
##  5          61 NEW CITY                     3    12.5          0.240
##  6           2 WEST RIDGE                   2     9.14         0.219
##  7           3 UPTOWN                       2     6.05         0.331
##  8           6 LAKE VIEW                    2     8.10         0.247
##  9           8 NEAR NORTH SIDE              2     7.12         0.281
## 10          10 NORWOOD PARK                 2    11.3          0.177
## 11          24 WEST TOWN                    2    11.9          0.169
## 12          30 SOUTH LAWNDALE               2    11.9          0.168
## 13          35 DOUGLAS                      2     4.27         0.468
## 14          42 WOODLAWN                     2     5.37         0.372
## 15          70 ASHBURN                      2    12.6          0.159
## 16          71 AUBURN GRESHAM               2     9.76         0.205
## 17           1 ROGERS PARK                  1     4.76         0.210
## 18           4 LINCOLN SQUARE               1     6.63         0.151
## 19           7 LINCOLN PARK                 1     8.20         0.122
```

```
## 20          12 FOREST GLEN                 1     8.28        0.121
## 21          13 NORTH PARK                  1     6.53        0.153
## 22          14 ALBANY PARK                 1     4.97        0.201
## 23          16 IRVING PARK                 1     8.33        0.120
## 24          17 DUNNING                     1     9.62        0.104
## 25          18 MONTCLARE                   1     2.56        0.390
## 26          19 BELMONT CRAGIN              1    10.1         0.0987
## 27          22 LOGAN SQUARE                1     9.30        0.108
## 28          26 WEST GARFIELD PARK          1     3.35        0.298
## 29          29 NORTH LAWNDALE              1     8.31        0.120
## 30          31 LOWER WEST SIDE             1     7.58        0.132
## 31          32 LOOP                        1     4.30        0.232
## 32          34 ARMOUR SQUARE               1     2.58        0.388
## 33          38 GRAND BOULEVARD             1     4.51        0.222
## 34          39 KENWOOD                     1     2.70        0.370
## 35          43 SOUTH SHORE                 1     7.60        0.132
## 36          44 CHATHAM                     1     7.65        0.131
## 37          45 AVALON PARK                 1     3.24        0.309
## 38          46 SOUTH CHICAGO               1     8.67        0.115
## 39          49 ROSELAND                    1    12.5         0.0801
## 40          51 SOUTH DEERING               1    28.2         0.0354
## 41          52 EAST SIDE                   1     7.73        0.129
## 42          53 WEST PULLMAN                1     9.23        0.108
## 43          54 RIVERDALE                   1     9.14        0.109
## 44          55 HEGEWISCH                   1    13.6         0.0737
## 45          56 GARFIELD RIDGE              1    11.0         0.0913
## 46          57 ARCHER HEIGHTS              1     5.20        0.192
## 47          58 BRIGHTON PARK               1     7.05        0.142
## 48          59 MCKINLEY PARK               1     3.66        0.273
## 49          60 BRIDGEPORT                  1     5.42        0.185
## 50          63 GAGE PARK                   1     5.69        0.176
## 51          64 CLEARING                    1     6.61        0.151
## 52          65 WEST LAWN                   1     7.65        0.131
## 53          66 CHICAGO LAWN                1     9.13        0.110
## 54          67 WEST ENGLEWOOD              1     8.17        0.122
## 55          68 ENGLEWOOD                   1     7.96        0.126
## 56          69 GREATER GRAND CROSSING      1     9.18        0.109
## 57          72 BEVERLY                     1     8.25        0.121
## 58          73 WASHINGTON HEIGHTS          1     7.40        0.135
## 59          74 MOUNT GREENWOOD             1     7.02        0.142
## 60          75 MORGAN PARK                 1     8.54        0.117
## 61          77 EDGEWATER                   1     4.50        0.222
## 62           5 NORTH CENTER                0     5.30        0
## 63           9 EDISON PARK                 0     2.94        0
## 64          11 JEFFERSON PARK              0     6.03        0
## 65          20 HERMOSA                     0     3.03        0
## 66          21 AVONDALE                    0     5.14        0
## 67          27 EAST GARFIELD PARK          0     5.01        0
## 68          33 NEAR SOUTH SIDE             0     4.62        0
## 69          36 OAKLAND                     0     1.57        0
## 70          37 FULLER PARK                 0     1.85        0
## 71          40 WASHINGTON PARK             0     3.94        0
## 72          41 HYDE PARK                   0     4.19        0
## 73          47 BURNSIDE                    0     1.58        0
```

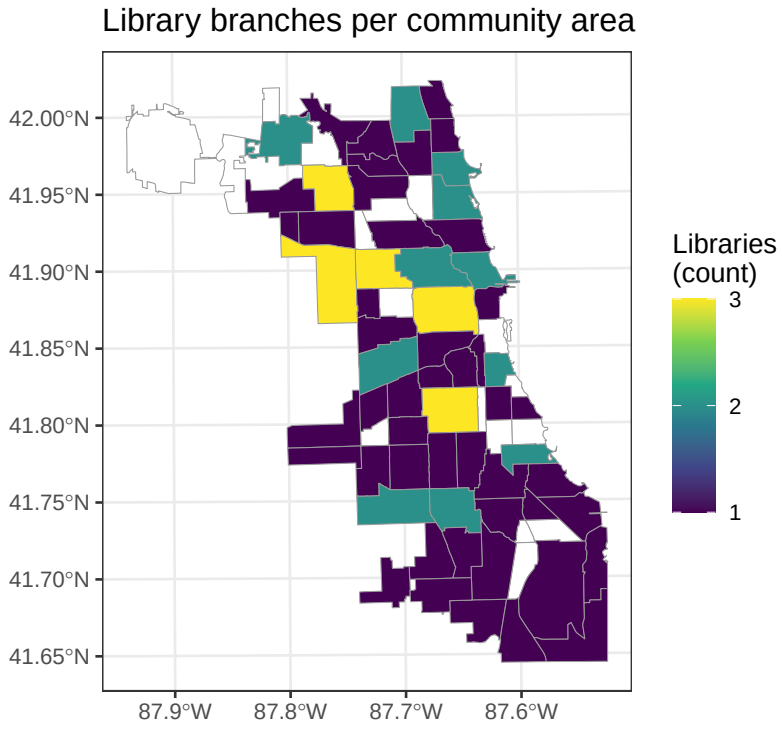
```
## 74          48 CALUMET HEIGHTS              0     4.54        0
## 75          50 PULLMAN                      0     5.48        0
## 76          62 WEST ELSDON                  0     3.04        0
## 77          76 OHARE                        0    34.5         0
```

```r
# Maps
map_caption <- paste0("Source: City of Chicago Data Portal (accessed 2026-11-04).\n",
                      n_unassigned, " sites left unassigned. White = 0 libraries, grey =
↪  missing.")
zero_areas <- filter(district_counts, n_sites == 0)
has_sites  <- filter(district_counts, n_sites > 0)

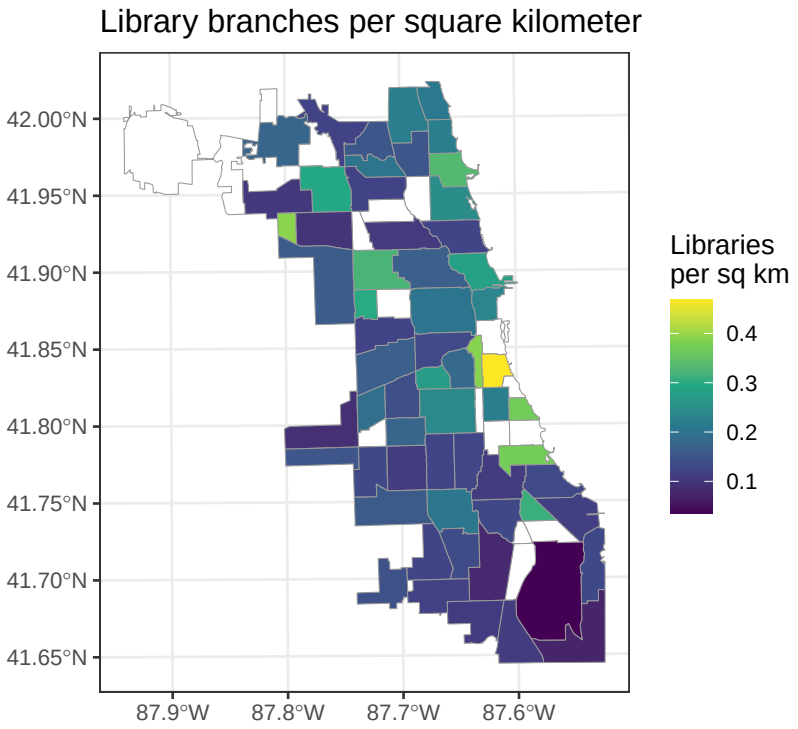
p_count <- ggplot() +
  geom_sf(data = zero_areas, fill = "white", color = "grey60") +
  geom_sf(data = has_sites, aes(fill = n_sites), color = "grey60") +
  scale_fill_viridis_c(name = "Libraries\n(count)", breaks = 1:3, na.value = "grey60") +
  labs(title = "Library branches per community area", caption = map_caption) +
  theme_bw()

p_density <- ggplot() +
  geom_sf(data = zero_areas, fill = "white", color = "grey60") +
  geom_sf(data = has_sites, aes(fill = sites_per_km2), color = "grey60") +
  scale_fill_viridis_c(name = "Libraries\nper sq km", na.value = "grey60") +
  labs(title = "Library branches per square kilometer", caption = map_caption) +
  theme_bw()

p_count
```



```
p_density
```



Source: City of Chicago Data Portal (accessed 2026−11−04).
0 sites left unassigned. White = 0 libraries, grey = missing.

```
# Boundary checks:
# 77 community areas, IDs 1 to 77 with names. No missing or duplicate IDs,
# no empty or invalid shapes, so no repairs or removals. Projected to EPSG:26971
# (meters), the same CRS as sites_m. All 77 areas are kept, including those with
# no libraries. Boundaries were last updated 2025-04-22 and libraries 2026-09-17,
# both from the City of Chicago, so they describe the same city geography.

# Matching audit:
# All 82 sites match exactly one area (0 sites with zero matches, 0 with multiple),
# so 82 sites are assigned and 0 are unassigned.
# Area checks: all areas are positive, and n_sites adds up to 82.
# 16 areas have 0 libraries, 45 have 1, 11 have 2, and 5 have 3.

# Interpretation:
# By raw count, the top areas are Portage Park, Humboldt Park, Austin, Near West Side
# and New City, with 3 libraries each. Accounting for area size changes the ranking:
# the highest densities are in small areas such as Douglas (0.47 per sq km), Montclare,
# Armour Square, Woodlawn and Kenwood, which have only 1 or 2 libraries. Austin has 3
# libraries but is large (18.5 sq km), so its density (0.16) is only middling. Large
# areas with no library, like O'Hare (34.5 sq km), look the same as small ones on the
# count map, so the density map gives a fairer comparison.
```